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SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:
Learn all about Landsat 9 in Space Age. Then in Earth, check out a jetfuel alternative; Solar Kerosene. Finally in R&D, find out how pearls get their shape

Source: Sciencenews.org

Source of earth's heat, trapped in a diamond

Researchers have discovered a mineral, which they named davemaoite, found inside diamond's flaws, which could be one of the answers to the age-old questions about the origin of heat inside the planet. https://dgit.in/hotmond

WHAT'S NEW

Changing climate and shrinking birds

There have been enough talks about how climate change has been affecting all life forms on earth, and if left uncontrolled, it could make the planet inhabitable. Well, another effect of climate change may be that the tropical birds around us may be getting smaller in size. In a recent study done in the untouched parts of the Amazon rainforest, it was revealed that over the last 40 years, the birds of the region had lost close to 2 per cent of their body weight.

Scientists have speculated that this loss of mass may be linked to the rising temperatures and how a smaller body would help the animals of all species shed heat off their bodies in a better and faster manner. This trend has also been observed in North American migratory birds, which have reportedly become smaller. These changes have coincided with the rise in the earth's temperature, further strengthening the theory that climate change may be causing the shrinkage. The average temperature rise in the Amazon was found to be 1 degree in the normal season and 1.67 degrees in the rainy season, at the same time, in the birds understudy, the motmot lost about 6 grams of its body weight. **https://dgit.in/birdshrink**

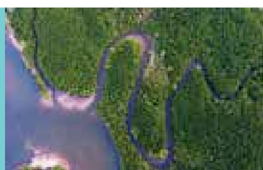
Source: ScienceNews

Fastest acceleration in the human body? It's a snap!

The snap that we execute using the fingers of our hands has been found by researchers to be the fastest accelerating action that the human body can perform. By deploying a series of high-speed sensors and 'state of the art' motion sensors, the scientists found out about the physics that goes behind the execution of a snap.

The reason behind this research is as fascinating as the findings.

The team of researchers, as told, had gotten into a heated debate about whether, while wearing the fabled 'Infinity Gauntlet', could Thanos snap or not. The famous snap from Avengers: Infinity War is what triggered the start of the study. Then careful observation techniques were used, and the findings revealed that the snapping action of fingers gives the fingers a maximum rotational velocity of about 7,800 degrees per second, and the maximal rotational acceleration is 1.6 million degrees per second squared. The finding also established that it would indeed have been physically impossible for Thanos to snap while donning that gauntlet. **https://dgit.in/fastsnap**

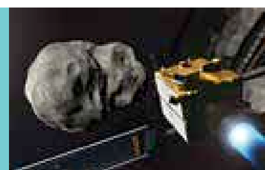


Satellites could fix ecosystems

Using spectrometers tied to aeroplanes and satellites to image the areas of jungles, a plant ecologist from the University of Minnesota, Jeannine Cavender-Bares, has figured out a way to fix the state of forest ecosystems and take corrective action.
https://dgit.in/satfix

Synthetic tissues to the rescue

Scientists from McGill University have figured out how, by using the combined knowledge of physics, chemistry, biology and engineering, they can develop synthetic tissues that could be used to repair human body parts like hearts, muscles, and vocal cords.
https://dgit.in/synthtis



Source: Nature.com

NASA Spacecraft for earth's defence

In a first, a spacecraft, launched by NASA, under their DART mission, is set to make a manoeuvre and clash with an asteroid, named Dimorphos, to figure out how, in the future, space crafts could be used for earth's defence from such bodies.
https://dgit.in/nasadef